Building Intelligent Applications With Spring AI

- Preface
  - Free benefits with your book
- Part 1: AI From the Beginning
- Chapter 1: Getting Started with Spring AI
  - Technical requirements
  - From start.spring.io to Spring Boot: Run in two minutes
  - Switching AI provider models in a single step
  - Summary
- Chapter 2: Understanding Spring AI: Part 1
  - Technical requirements
  - A brief overview of AI
  - Core APIs
    - ChatModel and ChatClient APIs
    - ChatModel API
    - ChatClient API
      - ChatClient auto-configuration
    - Advisors API
      - Spring AI provided advisors
      - Custom-crafted advisors
  - Moderation model API
  - Embeddings model API
    - Embeddings, the SimpleVectorStore and similarity search
  - Image model API
  - Audio model API
  - Summary
- Chapter 3: Understanding Spring AI: Part 2
  - Technical requirements
  - Synchronous vs asynchronous calls
  - Prompts, Prompt Templates and Messages
    - Prompts
    - Prompt templates
    - Prompt engineering
  - Structured output

The Spring AI BeanOutputConverter
Handling documents
Calling functions
StockQuotesFunction application
Spring AI API and AI provider reference
Summary
Part 2: Practical Spring AI
Chapter 4: Real-Time Message Translation
Technical requirements
Real-Time language translation for messaging
Application requirements
Code generation
Language translation using AI
Audio transcription (Speech-to-Text)
Text-to-Speech
Summary
Chapter 5: Beat Shazam
Technical requirements
Backstory
"Out of the sky, into the dirt"
Implementing Shazam
Processing audio with the Java Sound API
Embedding audio using Spring AI
Identifying a song using Spring AI
Running Shazam
Summary
Chapter 6: Connecting the Dots
Technical requirements
Implementing Connect 4
Prompts and applied features of Spring AI
Prompts and structuring outputs
The game board data structure
Explanations and AI model reasoning
Summary
Part 3: Beyond Simple Intelligence
Chapter 7: Become a Spring AI Power User
Technical requirements
Using chat and model options
Top-K and Top-P
Example of using Top-K and Top-P

- Tying up
- Metadata
  - Using the Metadata API
  - TokenCountEstimator
- Observability
  - Observing Token Count
  - Observing latency
  - Observing Call Count
- Testing
  - Use evaluation testing
  - Use discrete measurements
- Use pre-generated answers.
- Extensions
  - Enabling a ChatClient with a composable ChatModel
  - DecoratedSimpleVectorStore
  - Custom text splitters
    - DocumentTextSplitter
    - NewlineTextSplitter
    - ParagraphTextSplitter
    - SentenceTextSplitter
    - Custom StructuredOutputConverters
- Other extensions
- Tools
- Summary

Chapter 8: Secret Agents

- Technical requirements
- Overview of agentic AI
- Building AI agents with Spring AI
- Invoking Spring AI agents using an MCP client
- Building AI agents with Embabel, Koog and LangChain4j
- Summary

Chapter 9: Demystifying the Impact of AI

- From past to present
- The disruptive force of AI
  - A history of disruption
  - Progression of change
- Coding with AI
  - Code generation
  - Beyond the code
  - Electrolytes, it's what AI needs

Lean, mean, fighting machine
Only as strong as your weakest link
The ethical dilemma
Just because we can doesn't mean we should
Zero-sum game
What's next
Chapter 10: Unlock Your Exclusive Benefits
Unlock this Book's Free Benefits in 3 Easy Steps
Index

Previous chapter
< Index